Artificial Intelligence based Radar Technology: Current Advancements, Future Directions and Challenges

Dr. Tapas Kumar Pal
Directorate of Radar Seekers and Systems
Scientist, Research Centre Imarat, DRDO
Hyderabad, India
tkp-007.rci@gov.in

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***Abstract*—The paper explores the Artificial Intelligence (AI) based Radar technology advancement and future directions and challenges way ahead. The integration of Artificial Intelligence (AI) into radar technology enhances the radar adaptability, efficiency and precision across various field of applications such as defence, autonomous vehicle and weather monitoring etc. The conventional radar systems which mainly rely on the deterministic signal processing technique for target detection. This paper emphasis on the evolution of radar technology with integration of Artificial Intelligence (AI) and Deep learning (DL) for advanced signal processing, precise target identification, dynamic clutter rejection and adaptive system design. AI enabled radar will revolutionize the defence applications in precise target classification, electronic warfare and enhanced surveillance. Emerging technologies such as Quantum radar and AI-enhanced edge computing will further achieve an unparalleled sensitivity and detection capabilities, particularly for stealth targets. Despite the unparalleled trajectory of advancements of AI Radar still persist some technical challenges in present scenario including data scarcity, computational complexity and necessity for robust explainable AI models to ensure reliability in mission critical applications. In near future AI driven cognitive radar, hybrid learning approaches and scalable AI solutions will offer promising directions for advancing futuristic radar technology.**

I. Introduction

Radar (Radio Detection and Ranging) technology is a crucial tool in many domains, ranging from defence to weather monitoring, navigation, and automotive systems. Its fundamental operation relies on the transmission and reception is used to calculate the range of objects [1]. While radar systems have historically been based on signal processing techniques and manual interpretation. Incorporation of Artificial Intelligence (AI) promises to significantly enhance radar capabilities, making it more adaptive, efficient and autonomous.

AI-based radar systems leverage Machine learning (ML) and Deep leaning (DL) models to process large datasets in real time, enabling these systems to:

- Detect and classify targets with greater accuracy
- Adapt to dynamic environments
- Reduce clutter and interference from unwanted signals
- Optimize system parameters like beamforming, transmission power and frequency

A. Radar Technology Fundamentals

Radar systems operates by sending electromagnetic waves (usually radio waves) and analyzing the reflected signals that return after encountering a target. The basic components of a radar system include the transmitter, receiver, antenna, and signal processor. The transmitter generates radio waves, which propagates through space and reflect off objects. The receiver then captures the reflected signals, and the signal processor extracts information such as the range, velocity and direction of the target [1]–[4]. The radar equation defines the relationship between transmitted power and received signal strength. This relationship is essential for understanding the limits of radar detection and performance:

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4 L}$$

where P_r: Received Power, P_t: Transmitted Power, G_t, G_r: Gains of the transmitting and receiving antennas, λ: Wavelength of transmitted signal, σ: Radar Cross Section (RCS) of the target, R: Distance to target, L: System Losses. This equation highlights that radar performance depends significantly on the distance to the target (as R^4 grows rapidly with range), target properties (RCS) and system design (transmitted power, antenna gain and wavelength).

B. Signal Processing in Conventional Radar Systems

In traditional radar systems, signal processing is performed using a sequence of algorithms that extract useful information from raw radar returns. Key stages include:

- **Pulse compression:** Improves range resolution by shortening the received pulse, achieving a finer target resolution.
- **Doppler Shift Analysis:** Measures the change in frequency caused by the relative motion between the radar and target. This shift provides information about the target's velocity.
- **Clutter Rejection:** Filters out unwanted signals from stationary or slow moving objects, allowing radar systems to focus on moving targets.

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These classical methods though effective, are often limited in their ability to adapt to complex, cluttered and dynamic environments. They rely heavily on predefined models and assumptions that are not flexible in handling new or unpredictable conditions.

II. Integration of AI into Radar Technology

The integration of Artificial Intelligence (AI) into Radar system has significantly enhanced its ability to detect, classify and track targets in a complex and dynamic environment. Using machine learning (ML) and deep learning (DL) algorithms, AI-based radar systems can adapt to changing scenarios, process large volumes of data and make real-time decisions with minimal human intervention [5]–[9]. The following sections will describe the details of AI's role in radar signal processing, target classification, clutter rejection and adaptive radar system design.

A. AI in Radar Signal Processing

Classical radar signal processing methods mainly relied on deterministic algorithms such as Fourier transforms and Kalman filters for target detection, clutter suppression and noise reduction. Although it is effective in many scenarios but these techniques struggled in high dynamics and high clutter environments. Artificial intelligence (AI) has introduced data-driven models that can adapt to high dynamic and complex environments to improve radar performance [10], [11].

1) AI-driven Target Detection: AI-based target detection involves identifying returns from intended targets, e.g. reflections from aircraft, vehicles or drones from the radar echo. Deep learning models, such as Convolutional Neural Networks (CNNs) are particularly effective in analyzing radar data, including spectrograms and Range-Doppler (RD) maps [12]–[14]. Let X represent the radar data matrix, where x_{ij} corresponds to the return signal at range i and Doppler frequency j. A CNN process X as follows:

$$Y = \text{softmax}(f(X, \theta))$$

where $f(X, \theta)$ is the CNN with parameter θ and Y is the output probability vector indicating the likelihood of different targets (e.g., "vehicle," "aircraft," "drone").

2) Noise Filtering with AI: Traditional filters, such as average or Weiner filters, operate under fixed assumptions about noise characteristics. AI, on the other hand, learns the statistical properties of noise and clutter from training data, enabling adaptive noise suppression [15], [16]. For instance, Auto-encoders Unsupervised Neural Networks designed for feature extraction can be used to filter radar noise. The encoder maps radar data X to a latent representation:

$$Z = f_{\text{encoder}}(X)$$

The decoder reconstructs the de-noised signal:

$$\hat{X} = f_{\text{encoder}}(Z)$$

By training on clean radar signals, the auto-encoder learns to suppress noise components while preserving target related features.

B. AI for Target Classification

Target classification in radar system involves distinguishing between different types of objects, such as vehicles, aircraft, and drones. AI models excel in this task by leveraging labeled datasets and learning complex patterns in radar signals [17].

1) Machine Learning for Feature-Based Classification: Traditional ML algorithms, such as Support Vector Machines (SVMs) and Decision Trees, classify targets based on hand-crafted features like signal amplitude, Doppler velocity, and range [18]–[22]. Let

$$x = [x_1, x_2, \ldots, x_n]$$

represent a feature vector derived from radar signals. An SVM seeks to find a hyper-plane defined by

$$w^T x + B = 0$$

where w is the weight vector and B is the bias term. The SVM maximizes the margin between target classes, ensuring robust classification.

2) Deep Learning for End-to-End Classification: Deep learning eliminates the need for manual feature extraction by learning hierarchical representations directly from radar data. For example, a Recurrent Neural Network (RNN) can be used to classify moving targets by analyzing temporal radar data [23]–[27]. Let X_t denote the radar input at time t. The RNN updates its hidden state as follows:

$$h_t = \sigma(W_x X_t + W_h h_{t-1} + B)$$

where W_x, W_h are weight matrices, B is the bias vector, σ is a nonlinear activation function and h_t is the current hidden state. The final output Y is computed using a softmax layer:

$$Y = \text{softmax}(W_y h_T + b_y)$$

where T is the length of the radar observation window and b_y is output bias.

C. Clutter Reduction Using AI

Clutter refers to unwanted radar returns from objects like buildings, trees, or ground surfaces. Traditional clutter suppression techniques, such as Moving Target Indicator (MTI) filters, are limited by their reliance on fixed assumptions about clutter motion. AI enables adaptive clutter reduction by learning the statistical properties of clutter and target signals [28], [29].

1) Clutter Modeling with Machine Learning: AI models can differentiate between clutter and target signals by analyzing their spectral or temporal characteristics. A supervised ML algorithm, such as Random Forests, can be trained to classify radar signals into clutter and non-clutter categories. Let x represent a feature vector derived from radar data. A random forest classifier predicts the class y as:

$$y = \text{majority vote}(\{h_m(x)\}_{m=1}^{M})$$

where $h_m(x)$ is the prediction of the m-th decision tree, and M is the total number of trees.

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2) Deep Learning for Adaptive Clutter Suppression: Deep Learning models, such as Generative Adversarial Networks (GANs), can synthesize clutter-free radar data [23], [24], [27]. A GAN consists of two networks: a generator G that synthesizes radar signals $\hat{X}$, and a discriminator D that distinguishes between real and synthetic data. The GAN is trained using the loss:

$$\min_G \max_D \left(\mathbb{E}_{X \sim p_{\text{data}}}[\log D(X)] + \mathbb{E}_{\hat{X} \sim G(z)}[\log(1 - D(\hat{X}))] \right)$$

where X is real radar data, and $\hat{X}$ is synthetic radar data generated from random noise z. The generator learns to produce realistic, clutter-free radar signals, which can be used for improved target detection.

D. Adaptive Radar Systems with AI

AI enables radar systems to adapt their parameters, such as beam direction, pulse repetition frequency, and bandwidth, based on environmental conditions and operational requirements [30].

1) Reinforcement Learning for Adaptive Beamforming: Reinforcement Learning (RL) allows radar systems to learn optimal beamforming strategies through trial and error [28]–[30]. The radar system interacts with the environment, receiving feedback in the form of rewards r. The RL objective is to maximize the expected cumulative reward:

$$\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]$$

where π is the policy (mapping states to actions) and γ is the discount factor. A deep Q-network (DQN) approximates the action-value function $Q(s, a)$ as:

$$Q(s, a) \approx f(s, a; \theta)$$

where s is the current state, a is the action, and θ represents the DQN parameters. The radar system adjusts its beamforming parameters to maximize $Q(s, a)$, improving target detection and tracking.

III. Applications of AI in Radar Technology

A. Autonomous Vehicles

In autonomous vehicles, radar is critical for detecting obstacles, pedestrians, and other vehicles, even in low-visibility conditions [31]. AI plays an essential role in:

- **Sensor Fusion:** AI algorithms combine radar data with LiDAR, cameras, and ultrasonic sensors to form a comprehensive 360-degree view of the vehicle's surroundings.
- **Object Tracking:** Machine learning models can track the motion of surrounding objects, predicting their future positions to avoid potential collisions.
- **Path Planning and Decision Making:** AI models predict the motion of detected objects and plan the vehicle's actions to ensure safe operation.

B. AI in Autonomous Radar Systems

A typical system might use a Kalman Filter for target tracking. The Kalman filter is a recursive algorithm that estimates the state of a system based on noisy radar measurements. Its equations are:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - H\hat{x}_{k|k-1})$$

where $\hat{x}_{k|k-1}$ is the predicted state of the system, K_k is the Kalman gain which weights the correction applied to the predicted state based on measurement uncertainty, z_k is the radar measurement at time k, and H is the measurement matrix that links the state to the measurements. Kalman filters are widely used for target localization and predictive tracking in autonomous systems.

C. Military and Defence Applications

Radar technology has long been a cornerstone in military operations, providing vital capabilities for surveillance, navigation, targeting and defence. The integration of Artificial Intelligence (AI) in radar systems has revolutionized the military radar operations, enabling unprecedented levels of precision, adaptability and autonomy. AI-based radar systems offer advantages over traditional methods, particularly in the areas of target identification and classification, electronic warfare, and surveillance [32]–[35]. Below, we will explore these applications in greater detail.

1) Target Identification and Classification: In modern military radar systems, identifying and classifying targets accurately is crucial for mission success. Traditional radar systems often rely on manual analysis or predefined algorithms that can struggle with differentiating between various objects, especially in cluttered or complex environments. AI-based radar systems, especially using Deep Learning (DL) techniques, have significantly improved the ability to classify targets [23]–[27]. One of the key approaches for target classification in radar is the use of Convolutional Neural Networks (CNNs). CNNs are a type of deep neural network designed to process structured grid data, such as images or spectrograms of radar returns. These networks can learn spatial hierarchies of features from the radar data, enabling them to recognize complex patterns and structures that may indicate specific types of targets (e.g., aircraft, drones, missiles, or even ground vehicles). Mathematical formulation for the CNN process includes the convolution operation in the form:

$$y(x, y) = (f * g)(x, y) = \sum_m \sum_n f(m, n)\, g(x - m, y - n)$$

where,
f represents the radar signal or spectrogram,
g represents the kernel or filter used to detect features (e.g., edges, shapes),
$y(x, y)$ is the output feature map after applying the filter.

Once trained, these networks can accurately classify radar returns into categories, allowing military personnel to quickly identify whether the signal corresponds to an enemy aircraft,

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a missile or another object of interest. In Air Defence systems, AI-based radar could automatically identify and classify incoming projectiles or hostile aircraft, reducing the response time needed for engagement. The system could distinguish between military and civilian aircraft, thereby preventing false alarms and ensuring the correct identification of threats.

2) Electronic Warfare (EW) and Countermeasures: Electronic Warfare (EW) refers to the use of electromagnetic spectrum-based technologies to interfere with or disrupt enemy radar systems. Enhanced EW radar systems enable more effective detection and jamming of enemy radar signals. AI-based systems can analyze the incoming radar signals in real time, identifying the threat and deploying countermeasures such as jamming or spoofing.

3) Signal Processing and Jamming:: AI models, particularly Reinforcement Learning (RL), can optimize the jamming techniques based on the behavior of the enemy's radar. The RL agent learns to adjust the jamming signal's parameters to effectively disrupt the enemy radar's ability to detect targets, while minimizing the risk of detection. The objective in a typical jamming scenario is to maximize the interference power while ensuring that the enemy's radar systems cannot identify the source of the jamming. The optimal jamming signal S_j can be formulated as:

$$S_j = \arg\max_{S_j} \text{Interference}(S_j) \quad \text{subject to } P(S_j) \leq P_{\max}$$

where S_j is the jamming signal, Interference(S_j) is a function measuring the effectiveness of the jamming signal, $P(S_j)$ is the power of the jamming signal and $P_{\max}$ is the maximum allowable power.

In addition to jamming, AI can also assist in spoofing where false radar signals are introduced to mislead enemy radar systems. The system could generate fake targets or decoys that mimic real objects to divert attention from the actual assets. In a conflict scenario, an AI-powered radar system could detect an enemy radar signal, assess its characteristics and choose the most effective countermeasure (e.g., electronic jamming or deception) to neutralize the threat. Over time, the AI learns to refine its responses based on past successes and failures.

4) Surveillance and Tracking: AI significantly enhances radar-based surveillance systems by improving the accuracy and reliability of target tracking and motion prediction. Traditional radar tracking systems, such as Kalman Filters or Extended Kalman Filters (EKF) often assume linear motion models, which may not always be applicable in the complex scenarios encountered in military operations [36]. AI models, especially Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks and Attention Mechanisms, are particularly well-suited for these tasks. These models can handle sequential data (like radar returns over time), learning patterns of motion and predicting future positions of moving targets, even in the presence of noise. The tracking of targets can be formulated as a state estimation problem where the target's position and velocity are estimated over time. AI-based systems can improve tracking by incorporating contextual information such as environmental conditions, historical movement data and even interactions with other entities (e.g., aircraft following evasive maneuvers). An AI-driven radar system deployed on a naval vessel could track multiple incoming missiles or enemy aircraft simultaneously. The system would use machine learning to predict the likely path of each target, enabling the vessel's defence system to engage the targets efficiently.

D. Weather Prediction and Monitoring

AI improves weather radar systems by enabling more accurate detection of weather phenomena, such as tornedoes, hurricanes and micro bursts. Machine learning models can process radar data to predict weather events in real-time, improving the lead time for weather warnings.

IV. Future Directions and Challenges

A. Quantum Radar and AI

Quantum radar represents the next-generation radar technology that promises enhanced sensitivity and the ability to detect targets where classical radar systems fail. Quantum radar relies on the principles of entanglement and superposition of quantum mechanics to improve the detection capabilities beyond traditional radar systems [37]–[42]. Quantum radar can exploit quantum correlations to detect objects with higher precision and in the presence of clutter or stealth technologies. The integration of AI into quantum radar systems further enhances the capabilities of smart signal processing, target classification and adaptive decision-making.

1) Principles of Quantum Radar: In a traditional radar system, electromagnetic waves is emitted and reflected back by the object. The radar system measure the time taken for the signal to return and its frequency shift to detect the object. However in a quantum radar, entangled photons are used instead of classical electromagnetic waves. One key advantage of quantum radar is the ability to use quantum entanglement to correlate the transmitted and received signals. This entanglement allows the receiver to analyze the properties of the reflected signal more accurately and enhancing the radar's ability to detect weak or distant targets. In the quantum radar model, the quantum state of the transmitted signal can be represented as:

$$|\psi_{\text{out}}\rangle = \alpha\,|\text{vacuum}\rangle + \beta\,|1\rangle.$$

where,
α and β are complex coefficients that determine the probability amplitude of the quantum state,
$|\text{vacuum}\rangle$ represents the vacuum state (no photon),
$|1\rangle$ represents a single-photon state.

2) Applications of Quantum Radar in Defence and Beyond: Quantum radar is very useful in detecting low radar cross section objects such as stealth aircraft or submarines etc. In addition, stealth technology often exploits radar's inability to detect small and highly reflective targets. The enhanced

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sensitivity of quantum radar can significantly improve detection capabilities. In Defence applications, quantum radar can provide critical advantages for missile Defence systems, as it can detect stealth missiles that conventional radar systems would miss. By improving target detection accuracy, quantum radar systems can offer greater reliability and response times for Defence operations [43]. A quantum radar system deployed on a military vessel could detect a stealth aircraft attempting to penetrate airspace or a submarine operating beneath the surface. The AI-powered quantum radar system would classify the detected object and assess its threat level and depending on the situation trigger countermeasures if necessary. Quantum radar is also being explored for civilian applications such as airport security, weather monitoring and environmental sensing.

B. Data Security and Synthetic Data Generation

Training AI models requires large, high-quality datasets, which are often difficult to acquire for radar systems due to the complexity of the environment. Techniques like synthetic data generation and data augmentation are being developed to generate realistic radar datasets for model training.

C. Robustness and Generalization

AI models in radar systems must be robust to environmental changes, sensor noise and adversarial attacks. Research techniques like adversarial training, uncertainty quantification, and explainable AI (XAI) will enhance the reliability and interpretability of radar systems.

V. Challenges in AI-Radar Integration

A. Data Scarcity and Labeling

AI models require large datasets for training. However, radar datasets are often proprietary and contain imbalanced samples. Semi-supervised learning and synthetic data generation address these challenges:

$$\min_{\theta} \sum_{i=1}^{n} L(f_\theta(x_i), y_i) + \lambda R(\theta)$$

where $R(\theta)$ is a regularization term encouraging generalization.

B. Computational Complexity in Real-Time Systems

Radar systems demand real-time performance, but AI introduces computational overhead. Optimization techniques include:

1) **Model Quantization:** Reducing precision of weights:

$$\omega_q = \frac{\text{round}(\omega \cdot 2^n)}{2^n}$$

2) **Edge Processing:** Deploying AI on dedicated hardware like FPGAs or GPUs to reduce latency.

C. Explainability and Trustworthiness

Radar AI must be interpretable, especially in defence applications. Explainable AI (XAI) frameworks, such as saliency maps, help visualize radar features:

$$\text{Saliency}(x) = \frac{\partial f(x)}{\partial x}$$

VI. Future Trends in AI for Radar Technology

A. Cognitive and Quantum Radar with AI

Cognitive radar dynamically adapts its transmission and processing:

$$\hat{x} = \arg\max_{x} E[R(x, y)]$$

where R is a reward function optimizing radar performance.

B. Quantum Radar

Quantum radar uses entangled photons for higher sensitivity. AI processes quantum states, improving noise discrimination:

$$\rho = |\psi\rangle\langle\psi|$$

where ρ and $|\psi\rangle$ represents the quantum state and wave function respectively. AI deciphers faint target echoes amidst quantum noise.

C. AI-Enhanced Edge Computing in Radar

Edge computing localizes AI processing at the sensor level. AI models like MobileNet are optimized for deployment:

$$\min_{\theta} \text{Latency}(\theta) + \lambda \text{Accuracy}(\theta)$$

D. Evolving AI Models for Radar Systems

Future models will combine multiple paradigms:

1) Hybrid Learning: Combining supervised and reinforcement learning.
2) Adversarial Training: Improving robustness against jamming and adversarial inputs:

$$\min_{G} \max_{D} \left(\mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]\right)$$

VII. Conclusion

The integration of Artificial Intelligence (AI) into Radar technology makes a paradigm shift, enabling radar systems to operate with unprecedented adaptability, intelligence and precision capability. This transformation is especially more evident in military and defence applications where AI enhances target classification, countermeasure deployment and surveillance operations. The potential of AI-driven radar systems will further be enhanced by emerging technologies such as Quantum radar and AI-powered edge computing, promising significant advancements in detecting and tracking targets under very dynamic conditions. However, the successful deployment of AI-enhanced radar systems requires addressing challenges such as enough data availability for model training, computational overhead and adversarial robustness. Efforts to integrate explainable AI and develop robust algorithms will be critical in ensuring the reliability and trustworthiness of these systems. The continued evolution of AI technologies coupled

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with advancements in Quantum radar and Edge computing, will redefine sensing and detection capabilities across military, defence, automotive and environmental domains paving the way for intelligent, scalable, and autonomous solutions in the near future.

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